Leah E. McTigue
Leah.McTigue@colostate.edu

Education:

University of Arkansas, USGS Cooperative Research Unit – Graduated August 2023

- Master of Science in Biology/Wildlife Science
- GPA: 4.0
- Key Courses: Invasion Ecology, Biometry, Wildlife Management and Conservation, Spatial Analysis in GIS, Introduction to R, Professional Development seminar, Spatial Ecology Seminar, Population Ecology, Conservation Biology

Michigan State University, College of Agriculture and Natural Recourses – Graduated May 2017

- Bachelor of Science in Animal Science with a concentration in Exotic and Companion Animal Biology and Management
- GPA: 3.54
- Key Courses: Ecological aspects of Animal Behavior, Animal Nutrition, Animal Behavior, Environmental Physiology

Publications and Fellowships:

- McTigue, L.E., Steel, Z.L. *in Review*. Resilience of Six Common Mammals to Megafires in the Southern Rocky Mountains. Fire Ecology.
- McTigue, L. E., M. van den Bosch, H. M. Boone, et al. 2026. "Global Trends in Drought Impacts on Wildlife—A Review." Global Change Biology32, no. 2: e70752. https://doi.org/10.1111/gcb.70752
- Rooney, B., Kays, R., Cove, M. V., Jensen, A., Goldstein, B. R., Pate, C., ... & Lesmeister, D. B. 2025. SNAPSHOT USA 2019–2023: The First Five Years of Data From a Coordinated Camera Trap Survey of the United States. Global Ecology and Biogeography, 34(1), e13941. https://doi.org/10.1111/geb.13941
- McTigue, L. E., DeGregorio B. A. (2024). Nine-banded Armadillo (*Dasypus novemcinctus*) Occupancy and Density Across an Urban to Rural Gradient. Southeastern Naturalist. 23 (4), 529-548. https://doi.org/10.1656/058.023.0413
- McTigue L. E., DeGregorio B. A. 2023. Effects of Landcover on Mesocarnivore Density Along an Urban to Rural Gradient. Global Ecology and Conservation. 48:e02716.https://doi.org/10.1016/j.gecco.2023.e02716
- McTigue L.E., Lassiter E.L., Shaw M., Johansson E., Wilson K., DeGregorio B.A. 2023. Does daily activity overlap of seven mesocarnivores vary based on human development? PLOS ONE. 19(1): e0288477. https://doi.org/10.1371/journal.pone.0288477
- Montgomery, T. M., Lehmann, K. D., Gregg, S., Keyser, K., McTigue, L. E., Beehner, J. C., & Holekamp, K. E. (2023). Determinants of hyena participation in risky collective action. Proceedings of the Royal Society B, 290(2011), 20231390. https://doi.org/10.1098/rspb.2023.1390
- Veon J., Lassiter E.V, Johansson E., Shaw M, McTigue L.E., Massey A., Gibson R., DeGregorio B.A. (*in review*) Influence of human development and predators on pattersn of Virginia opossum occupancy, abundance, and activity. Journal of Zoology. https://doi.org/10.1111/jzo.13111
- Harry and Jo Leggett Chancellor's Fellowship, awarded fall 2021 (Leggett Fellowship)

Relevant Experience:

Ecological Research Associate, Colorado State University *August 2023-present*

- Assess the impact of burn severity and pyrodiversity on wildlife communities.
- Place survey equipment (game cameras and automated recording units) at three burn units from 2020 - Cameron peak, East Troublesome, and Mullin fires.
- Plan fieldwork and deployment schedules across burn units

- Identify species from recordings and trail camera photographs through Timelapse 2.0 software, birdNET, and Sonobat.
- Hire and manage field technicians
- Equipment purchasing and budget management
- Data management and manipulation
- Statistical analysis
- Collaborate with fellow researchers on various projects
- Vegetation sampling and identification

Master's Thesis, University of Arkansas at Fayetteville *August 2021 – July 2023*

- Chapter 1: Mesocarnivore Density Along an Urban to Rural Gradient
    - Estimated density for 5 mesocarnivore species using the Random Encounter Model
    - Generalized Linear Mixed Models
- Chapter 2: Armadillo Occupancy Along an Urban to Rural Gradient
    - Single species occupancy analysis
- Coordinated 1 lab project as lead author resulting in submission to PLOS ONE
- Participated in lab projects resulting in 2 manuscripts, one submitted to Journal of Zoology, one in progress.
- Summarizing research into yearly reports
- Sorted 600,000+ photos using timelapse2.0 software
- Extensive use of Arc GISPro for mapping and data collection
    - Extraction of landcover and anthropogenic variables
    - Plotting of all camera locations
- Data entry, cleaning and organization using Excel and R Studio
- Statistical analysis using R studio
    - Generalized linear mixed models
    - Occupancy analysis using package unmarked
    - Overlap analysis using packages suncalc, and overlap

Elk Biological Aide, Pennsylvania Game Commission *September 2020-July 2021*

- Assisted with darting and processing of over 50 chemically immobilized elk
- Assisted with vegetation surveys for microscale habitat selection of elk
- Worked elk check stations registering harvested elk and collecting biological samples
- Monitored >50 GPS and radio collared elk for survival
- Maintained trail camera setups
- Assisted with vegetation and habitat surveys
- Completed routine data entry and administrative record keeping
- Vegetation sampling and identification
    - Grasses, forbs, woody shrubs, trees, etc.

Research Associate, Isle Royale National Park Wolf Project *June-September 2020*

- Completed wolf cluster investigations to determine activity and space use
- Assisted with the maintenance of a mid-sized camera trap array
- Completed visual obstruction vegetation surveys for the Isle Royale camera array
- Conducted scat transects and howl surveys for non-collared wolf monitoring
- Processed wolf scats for genetic, parasite, stress hormone and diet analysis
- Completed 1–4-night backcountry stints on Isle Royale National Park
- Completed 8-16 miles per day with up to a 60lb pack

- Set and maintained Acoustic Recording Units for wolf monitoring
- Completed telemetry surveys to monitor collared individuals

Wildland Fire Fighter, WDNR *March 2020*

- Assisted in prescribed burns in a mix of prairie and woodland ecosystems
- Operated hand tools including Polaskis, rakes, shovels, flappers for fire break creation and fire suppression
- Carrie and operated backpack water tanks (weighing up to 50 pounds) and hoses for fire suppression
- Hiking up to 10 miles while carrying tank packs for fire suppression in hot and smokey conditions
- Participating in mop up post prescribed burn.

Sample Technician, WDNR CWD Deer and Predator Project *September 2019-June 2020*

- Use of Chemical immobilization on White Tailed Deer and Bobcats
- Collected biological samples from deer
- Adult deer capture and restraint using drop nets and clover traps
- Baiting private properties with corn for drop nets and clover lines
- Used radio telemetry to detect, triangulate, and locate deer mortality events
- Conducted mortality investigations
- Used ATVs and UTVs in highly variable conditions
- Walked 2-8 miles per day carrying up to 50lbs
- Use of trail cameras to monitor deer patterns
- Maintained positive relations with private landowners
- Harvested lymph nodes from 100+ deer for CWD testing
- Tagging photos by species for the Snapshot Wisconsin Project

Mountain-lion Kill-site Technician, Utah State University *May-September 2019*

- Independently conducted 142 mortality investigations on active mountain lion kill sites
- Hiked 3-13 miles in steep, off-trail terrain, often in areas with a high predator density
- Use of trail cameras to monitor mountain-lion feeding behavior and the presence of other species at the kill sites
- Tagging species present at mountain lion kill site from videos in ViXeN
- Use of Topographic maps and handheld GPS device to navigate off trail
- Maintained accurate data sheets for kill sites
- Entered data sheets into Microsoft Excel
- Worked with local landowners to gain access to private property for mortality investigations
- Assisted with locating, collaring and ear tagging 5-week-old mountain lion cubs

Wildland Fire Fighter, WDNR *March 2019-May 2019*

- Assisted in prescribed burns in a mix of prairie and woodland ecosystems
- Operated hand tools including Polaskis, rakes, shovels, flappers for fire break creation and fire suppression
- Carrie and operated backpack water tanks (weighing up to 50 pounds) and hoses for fire suppression
- Hiking up to 10 miles while carrying tank packs for fire suppression in hot and smokey conditions
- Participating in mop up post prescribed burn.

Leah E. McTigue
Leah.McTigue@colostate.edu

Basic Technician, WDNR CWD Deer and Predator Project *November 2018-May 2019*

- Adult deer capture and restraint using drop nets and clover traps
- Baiting private properties with corn for drop nets and clover lines
- Used radio telemetry to detect, triangulate, and locate deer mortality events
- Conducted mortality investigations
- Used ATVs and UTVs in highly variable conditions
- Walked 2-8 miles per day carrying up to 50lbs
- Use of trail cameras to monitor deer patterns
- Maintained positive relations with private landowners
- Harvested lymph nodes from 100+ deer for CWD testing

Research Assistant , Holekamp Mara Hyena Project *July 2017-June 2018*

- Daily behavioral and demographic data collection on spotted hyenas
- Assisted with the collection of behavioral data and biological samples for specific graduate projects within the lab
- Used radio telemetry to track collared spotted hyenas
- Trained to handle and use a Telinject Air Rifle for the darting of hyenas
- Assisted with darting, workup, and post darting bloodwork of 8+ hyenas
- Conducted bi-monthly prey censuses
- monitored sympatric carnivore species
- Managed the field site, vehicles and staff
- Trained in new research assistants
- Worked with local Maasai and park management for community relations

Undergraduate Research Assistant, Holekamp Mara Hyena Project *October 2015-May 2017*

- Worked on an independent research project, to be presented at MSU University Undergraduate Research and Arts Forum, Spring 2017
- Entered qualitative data into Microsoft Access database for lion hyena interaction research
- Hyena Education Outreach team member, aiming to educate elementary school students about hyena conservation

Posters and Presentations

- McTigue, L.E., (March, 2026). Colorado Fires and Biodiversity. Rocky Mountain National Park Biennial Research Conference. Oral Presentation.
- McTigue, L. E., (October 2024). The Effects of Megafires on Mammal Communities in the Colorado Rocky Mountains. The Wildlife Society National Conference, 2024. Poster Presentation.
- McTigue, L. E., (August 2022). Mesocarnivore Density on an Urban to Rural Gradient in Northwest Arkansas. The Wildlife Society Arkansas Chapter Conference, 2023. Formal Presentation.
- McTigue, L.E., (November 2022). Mesocarnivore Density on an Urban to Rural Gradient in Northwest Arkansas. The Wildlife Society National Conference, Spokane, 2022. Poster Session.
- McTigue, L. E., (August 2022). Mesocarnivore Density on an Urban to Rural Gradient in Northwest Arkansas. U.S. Geological Survey, Arkansas Cooperative Fish & Wildlife Annual Cooperators Meeting.
- McTigue, L. E. (September 2021). Lightning presentation given at the U.S. Geological Survey, Arkansas Cooperative Fish & Wildlife Annual Cooperators Meeting.

- McTigue, L. E., Lehmann, K, Montgomery, T, Holekamp, K, (April 2017). Sex differences in participation in mobbing behavior by spotted hyenas (*Crocuta crocuta*) during lion-hyena interactions. Annual University Undergraduate Research and Arts Forum. Michigan State University. Poster Presentation.

Services

Peer Review

- Balme, N., Woodgate, Z., Hunter, L., & Balme, G. (2025). A struggle to survive: Guild hierarchy predicts drought benefit among large carnivores. *Journal of Animal Ecology*.Journal of Forest Ecology and Management
- Kuntze, C. C., Pauli, J. N., Keane, J. J., Roberts, K. N., Dotters, B. P., Kramer, H. A., & Peery, M. Z. (2025). Multi-scale forest heterogeneity promotes occupancy of dusky-footed woodrats in the Sierra Nevada. *Forest Ecology and Management*, *578*, 122412.
- DeGregorio, B. A., & Deshwal, A. (2025). Current Distribution of the Nine-Banded Armadillo (*Dasypus novemcinctus*) in the United States. *Diversity*, *17*(2), 138.

Honors Thesis Advisor

- Assisting a past technician with her honors thesis project
  - Assisting with the scientific process
  - Teaching and assisting with statistical analysis
  - Mentoring through the writing process and creation of a professional poster presentation, to be presented at the Celebrate Undergraduate Research and Creativity (CURC) Showcase

Other

- University of Arkansas – The Wildlife Society Student Chapter President, 2022/2023
  - Organized events for undergraduates interested in wildlife studies, including a radio telemetry workshop, assisting with bird banding, and a presentation/question session with an Arkansas Game and Fish Commission employee.

Skills and Certifications

- Wilderness First Aid and epinephrin administration Certification (May 2024, NOLS)
- Proficient in ArcGIS Pro, R, Excel, PowerPoint, Word, Access, GPS Plus, ArcGIS Maps
- North American and East African taxonomy, skull, bone, and track identification
- Chemical Immobilization certification (PA, WI)
- GPS, map, and compass orienteering
- Proficient using radio telemetry to monitor large mammals
- Excel in data organization and entry
- Able to drive manual or automatic 4WD vehicles on poorly maintained roads in variable conditions